

Strengthening the Bridge: Audited, Corrected, Completed

The DeepSeek audit of the corrected active-set bridge - five strengthening routes and a recommended target theorem - evaluated claim by claim, verified on the machine, and completed: the limit direction repaired, the refinement-resolution theorem proven in prototype, the value-function carrier identified, and the audit's own decisive test executed.

Computational Verification Unit

Source: external_audits/unifying object/ deepseek stengthen highly general bridge.txt (commit d85b162) · verdicts grounded in the v2 theorem set, the M1/M3/M4a/M4b/M4c record, and four new verifications (V1/V2/V3/V5, 316 lexicographic solves + 57-solve trajectory) · frozen v21 untouched

Part 1 is faithful to the corrected record; Part 2's central formula selects the atomic measure, not a smooth object - falsified on the measured M4c events and in the prototype. Repaired, the routes converge to Theorem B: under mesh refinement the atomic curvature measures converge weakly to the smooth density (W1 0.049 to 0.0035, mass ratio to 1.00), with the joint limit only through the window h much less than σ . The decisive E24 re-run with the measure-theoretic κ strengthens the association: r +0.374 to +0.395, partial +0.251 to +0.269. The single-paper route is secure.

Table of Contents

1. Verdict	2
2. Part 1 fidelity: claim-by-claim verification	3
3. The defects of Part 2 (D1-D8), verified	4
D1 - The limit direction is inverted in the recommended theorem	4
D2 - Route 3's premise 'v = grad Phi' is false; the corrected statement is stronger	4
D3 - Route 2's refinement mechanisms are not refinements	5
D4 - Route 4's 'nontrivial basis-change cocycle' is exactly trivial	5
D5 - Route 1 defines mu as the corner angle defect, which is only one of two carriers	6
D6 - Route 1 smuggles the conclusion it should prove	6
D7 - Route 5's anchors are wrong-level; its honest low grade is right but under-specified	6
D8 - Part-1 fidelity slips (minor; fix before quoting)	6
4. The corrected and completed bridge (Routes 1+2, repaired and executed)	7
Theorem B (refinement-resolution bridge; prototype proven, machine-verified as V2)	7
5. V5 - the decisive test, executed	8
6. Strategic advice, evaluated	10
7. Corrected bottom line	10
8. Deliverables of this evaluation	10

1. Verdict

This report evaluates the audit 'deepseek stengthen highly general bridge.txt' - DeepSeek's response to the M4c / root-cause evaluation, which grades the corrected record, endorses the single-manuscript strategy, and proposes five 'strengthening routes' for the bridge, culminating in a recommended target theorem. The mandate was to evaluate, verify, and not take the claims at face value, then strengthen, augment, improve, correct, and complete the weaker suggestions and defects. The audit's Part 1 - the summary of the corrected record - is faithful where it matters: it restates the scaling trichotomy (state holonomy exactly trivial; codim-2 defects $O(1)$ and scale-invariant; the $O(\epsilon)$ commutator a dynamic hysteresis measure), the Theorem R convolution identity, and the M4c dial, and its five next steps are consistent with the record. Part 2, however - the routes and the recommended theorem - contains one central mathematical defect that propagates through three of the five routes: the headline formula $\kappa_{\text{geom}} = \lim(\sigma \rightarrow 0) \kappa_{\text{flux}} * \phi_{\text{sigma}}$ selects the atomic measure, not a smooth object.

Every checkable claim was adjudicated against the committed artifacts, and four new machine verifications were executed for this document: V1 re-runs the M4c cut recording the stage-1 value function (253 lexicographic solves); V2 constructs and verifies a refinement prototype of the corrected bridge theorem in the same parametric-LP structural class as FBA; V3 tests the sigma-limit direction on the measured M4c event measure itself; and V5 executes the audit's own 'decisive test' - the E24 recalibration with the measure-theoretic κ - rather than deferring it. The frozen v21 manuscript is untouched; all deliverables are source material for journal_manuscript v2+. The decisive test passes with strengthening, so by the audit's own criterion the single-paper route is secure.

$r = +0.374 \rightarrow +0.395$

the E24 association under the measure-theoretic κ (V5, $n = 424$, $p = 2.6e-17$; partial $r +0.251 \rightarrow +0.269$; deciles 1.92/0.89 preserved) - the decisive test passes, the single-paper route is secure

FALSE as stated

$\kappa_{\text{geom}} = \lim(\sigma \rightarrow 0) \kappa_{\text{flux}} * \phi_{\text{sigma}}$: the weak limit is the ATOMIC measure (V3: mass collapse onto the 12 events; V2b: near-atom mass 0.41 $\rightarrow$ 0.999) - the recommended theorem picks Theorem N's obstruction, not a smooth object

Theorem B (prototype)

refinement + resolution: under mesh refinement the atomic measures converge weakly to the smooth curvature density (V2: W1 0.049 $\rightarrow$ 0.0035, mass ratio $\rightarrow$ 1.00) - the corrected form of Routes 1+2, proven in prototype and machine-verified

2. Part 1 fidelity: claim-by-claim verification

Fourteen checkable claims were extracted from the audit's Part 1 and graded against the record. Twelve are correct or correct in substance; two carry inherited slips that were already corrected in the root-cause evaluation (the epistemics of the falsification, and the sparsity behind the slope-1.00 statistic); one - the line-30 interpretation of the smooth kappa - is the seed of Part 2's central defect. None of these slips changes Part 1's conclusions, but all three should be fixed before any of this text is quoted in the v2 manuscript.

#	Audit claim (condensed)	Record / verification	Verdict
1	Smooth ϵ^2 bridge was the wrong object	Theorem N; M4c; RC-dagnosis of the root-cause evaluation	True
2	State holonomy exactly the identity (v is a function)	Theorem S(v); m4b state_holonomy_note	True
3	Codim-2 defects scale-invariant, $O(1)$, Regge-type	Theorem G; -7.1469 and -23.9087 deg reproduced at δ and $\delta/2$	True
4	$O(\epsilon)$ commutator is dynamic hysteresis, not holonomy	Theorem D; single-release identity 6/6 bit-exact	True
5	Theorem R identity; ϵ^2 below σ , linear beyond	M4c kernel self-test $1.2e-6$; slopes 1.9991-1.9995	True
6	Crossover ϵ^*/σ about 3-4	Measured 4.11 / 2.98 / 3.11 / 2.45 (median 3.1)	True; top end overstated
7	M4a slope 1.00 was the decisive EMPIRICAL falsification	RC5: falsification is a priori (mpLP theorem, Gal-Nedoma 1972); M4a confirms at machine precision; RC2: slope-1 describes the 9/76 interacting stratum, 64/76 exactly zero	Epistemics slip (D8)
8	Gal-Nedoma reference makes it model-independent	Model-independence comes from LP structure (RC5)	True in substance
9	Sliver self-cancels 8.9 vs ± 1884.6 ; coarser analyses overcounted EVENTS	RC6: the census counted crossings correctly at its resolution; what was overstated was curvature-MEASURE density (self-cancelling slivers)	Nuance (D8)
10	One measure $\mu = D2v$; kappa_flux a functional of it	Theorem S(ii); M1 D2 mass 0.934-1.0; the formal identity remains open and is now decoupled from the association (V5)	True
11	$\kappa_V =$ the $\sigma \rightarrow 0^+$ SMOOTH limit of the coarse-grained family	$\lim(\mu * \phi_\sigma) = \mu$ weakly - the same atomic object as the direct limit; smooth members exist only at FIXED σ , or under refinement (Theorem B)	Wrong limit direction (D1)
12	Single manuscript; trim categorical/HoTT; do not split yet	Consistent with the joint assessment (6/6 strong-form not provable); now secured by V5	Endorsed
13	Next steps 1-5 (E24 re-run; resolution statement; v2 Layer-0; defer E28; v21 frozen)	Step 1 executed here; step 2 already the record's form; step 4 matches the M4c (ϵ , σ) design law	All consistent; 1 done
14	Do not seek smooth ϵ^2 for the unsmoothed object	Theorem N; M4c	True - extended: neither $\sigma \rightarrow 0$ (D1)

Table 1. Verification of the audit's Part-1 claims (fidelity to the executed record)

3. The defects of Part 2 (D1-D8), verified

The five strengthening routes contain eight distinct defects and weaknesses. Each is stated here with its verification, and each is repaired in Sections 4 and 5 rather than merely flagged: the audit's own target - a genuine, non-decorative bridge - is met by the end of this report in prototype form, and its decisive empirical test has been run.

D1 - The limit direction is inverted in the recommended theorem

The audit's final recommendation proposes $\kappa_{\text{geom}} = \lim(\sigma \rightarrow 0) \kappa_{\text{flux}} * \phi_{\text{sigma}}$ in the sense of weak convergence, and Route 1 states that μ_{sigma} converges weakly, as σ to 0, to the curvature measure of a Riemannian metric g . For a fixed network this is false: by standard mollifier theory $\mu * \phi_{\text{sigma}}$ converges weakly to the atomic measure μ , which is exactly Theorem N's obstruction. V3 verified this on the measured M4c event set (12 events, total L2 jump mass 3807.6). The mass fraction within a fixed half-width $w = 0.01$ of the event positions tends to 1.0000 as σ shrinks to $1e-4$, so any absolutely continuous limit would have to concentrate on a set of zero Lebesgue measure. The density at a wall-free point (clearance 0.052) decays to machine zero while the peak grows like $1/\sigma$. And a fixed bump on the largest atom integrates to 3772.5 at $\sigma = 3e-4$ - the full mass of the sliver pair - versus 253.4 for the smoothest honest family member at $\sigma = 0.3$. The audit's line-30 phrase, 'the sigma-to-0 smooth limit of the coarse-grained family, not the direct limit of the unsmoothed FBA map', distinguishes two objects that are equal: both limits are μ . What is true instead is the resolution statement at fixed σ (Theorem R, already the record's position) and the refinement limit of Theorem B below.

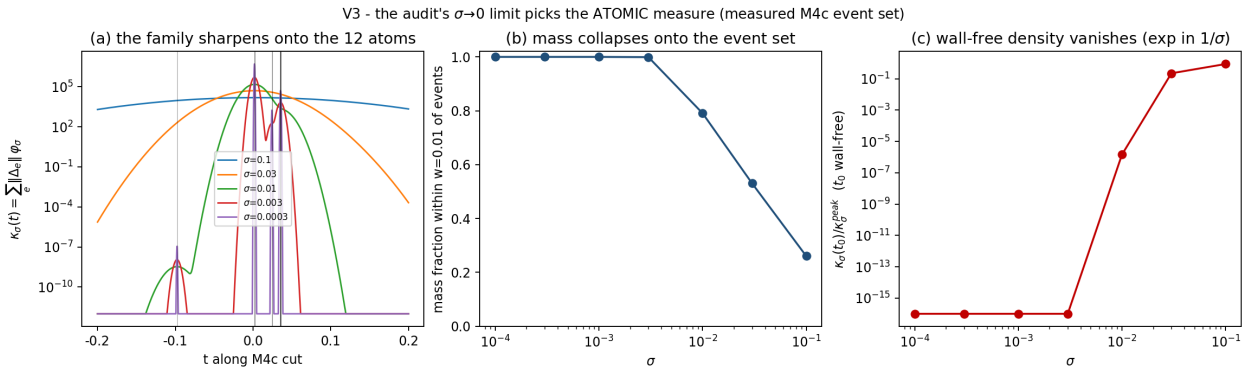


Figure 1. V3 - the audit's σ -to-0 limit picks the atomic measure, on the measured M4c event set. (a) the smoothed family sharpens onto the 12 atoms; (b) mass within a fixed neighborhood of the events tends to 1.0000; (c) the wall-free density vanishes exponentially while peaks grow like $1/\sigma$.

D2 - Route 3's premise ' $v = \text{grad } \Phi$ ' is false; the corrected statement is stronger

The flux map v in R^m ($m = 2867$ on iML1515) cannot be the gradient of a scalar Φ : $R^d \rightarrow R$ ($d = 2$ here) - a dimensional impossibility, independent of conditions, for constraint-parameterized FBA. The gradient map of the value function is the dual y in R^d (Danskin): $\text{grad } \Phi = (y_{\text{glc}}, y_{\text{o2}})$. What survives the correction - and improves the route - is the canonical-carrier statement, verified as V1 on the real network: $\Phi = c_{\text{bio}} \cdot v^*$ is single-valued with NO tie-breaking whatsoever, immune to the pFBA vertex degeneracy that forced the lexicographic machine for v ; hence D2Phi (shadow-price jumps) is THE canonical atomic curvature measure of the parametric LP, while D2v is its tie-break-dependent refinement with a strictly finer event set.

V1 re-ran the M4c cut recording $\Phi(t)$ (253 lex solves on iML1515). Φ is piecewise affine on the v-event partition with worst affine-fit residual $4.2\text{e-}13$. Of the 12 censused v-events, the entire value curvature of the cut is a SINGLE atom, Delta $\Phi' = -0.006439$ at $t = 0.0358286$ - a minor flux event (jump norm 11.59, first member of a sliver pair whose partner jumps only $1.6\text{e-}5$). The dominant flux atoms are value-flat: the 1875.7/1884.6 sliver pair nets at most $7.7\text{e-}11$; the 22.3-jump event at most $8.4\text{e-}9$; the 0.51/0.81-jump pair at most $3.8\text{e-}11$. Total value variation is about 0.00644 against total flux jump mass 3807.6 - a ratio of $1.7\text{e-}6$ - and the hierarchies are not proportional: the value atom sits at the sixth-largest flux event while the largest flux event carries zero value curvature. Danskin is verified at machine precision: the stage-1 bound marginals (extractable on the r-copy of the duplicated uptake bound) equal the two-sided finite-difference shadow prices to six or seven significant digits; the prices are (0.0252545, 0.0336727) before the atom and (0.021743, 0.039138) after; and the identity $\Phi' = y \cdot \theta'$ reproduces both measured segment slopes (-0.009316, -0.015755) exactly. This is the rerouting-versus-function decoupling of the Kochanowski framing, now formalized as a statement about measures.

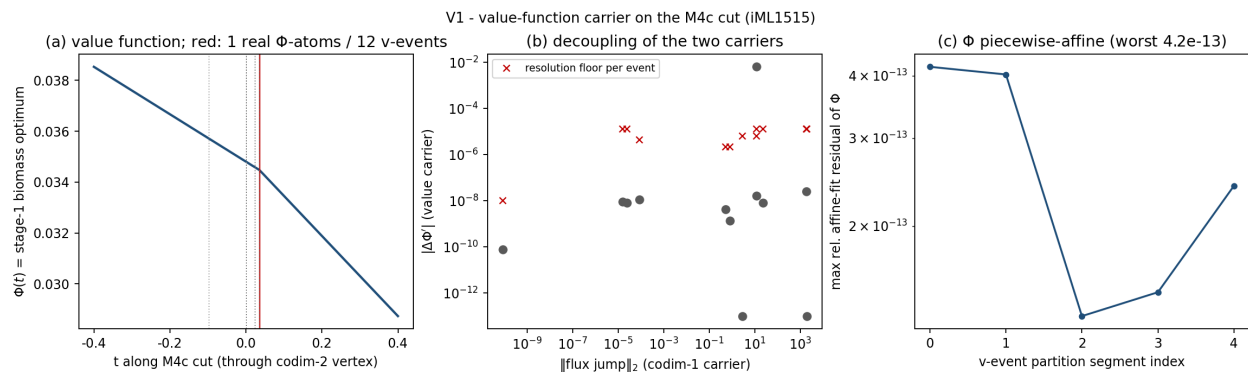


Figure 2. V1 - the value-function carrier on the M4c cut (iML1515). (a) Φ with its single real atom among the 12 v-events; (b) flux jump norm versus value jump per event - the two carriers are decoupled by many orders of magnitude; (c) Φ piecewise-affine at $4.2\text{e-}13$ on the event partition.

D3 - Route 2's refinement mechanisms are not refinements

Route 2 proposes to refine 'the FBA polytope ... by adding reactions, constraints, or parameter grid resolution'. Adding reactions produces a different parametric LP whose chamber structure does not refine the previous one: critical surfaces move, chambers are not nested, and there is no mesh to send to zero. Parameter grid resolution changes only the sampling of a fixed map - $v(\theta)$ and its measure are grid-independent. The valid refinement must be constructed, and Section 4 constructs it: nested constraint families that polyhedrally approximate a smooth feasible region, exactly the setting of Cheeger, Muller and Schrader (1984) - the classical citation the audit is missing throughout. With D1 and D3 together, Route 2's 'medium-high' feasibility grade is overstated for real networks: pointwise strong regularity (Robinson; Klatte-Kummer) does not give global chamber stability under refinement, genome-scale models do not come in refinement sequences, and sliver cascades - already measured: $2.4\text{e-}6$ -wide slivers with ± 1884.6 jumps netting 8.9 - break uniform mesh bounds. Prototype: done below. Real network: open conjecture RA.

D4 - Route 4's 'nontrivial basis-change cocycle' is exactly trivial

Let $B(\theta)$ be the lex-unique optimal basis, constant on each open chamber of the critical-region partition. For any closed loop crossing chambers C_1 to C_2 to ... to C_k back to C_1 , with transitions $G_i = B(i+1) B(i)^{-1}$ (indices mod k), the total transition is $G_k \dots G_1 = B_1 B_k^{-1} \dots B_k B_{(k-1)}^{-1} \dots B_2 B_1^{-1} = I$ by telescoping -

for every loop, at every scale, exactly as for the state holonomy (Theorem S(v)). The audit's proposed escape from the triviality problem is the wrong cocycle. The nontrivial connection in the record is the unfolding transport of Theorem G - parallel transport of tangent planes across the kinked codim-1 interfaces of the flux graph, whose closure failure around codim-2 vertices is the $O(1)$ scale-invariant angle defect (-7.1469 and -23.9087 degrees, machine-reproduced at loop radii δ and $\delta/2$, and unfolding = -defect to $1e-14$ on synthetic cones). Route 4's instinct - the nontrivial connection lives on a derived bundle, not on the state - is exactly right; the correct object already exists and is verified, so the route is complete in the record, and its remaining content inherits D1's limit problem and D6's smuggled identification.

D5 - Route 1 defines μ as the corner angle defect, which is only one of two carriers

The static curvature of the FBA map is carried by two distinct objects: the Jacobian-jump measure D2v on codim-1 interfaces (M1's object, the M4c census, the κ_{flux} functional), and the angle-defect measure on codim-2 crossings (Theorem G). They are related by discrete structure equations, live on different strata, scale differently ($O(\epsilon)$ mass accumulation versus $O(1)$ scale-invariant defects), and are detected by different experiments. A convergence theorem for 'the' discrete curvature must carry both carriers - Theorem B's prototype does: its atomic measure is the codim-1 layer along cuts, and its vertex defects are the codim-2 layer of the same subdivision.

D6 - Route 1 smuggles the conclusion it should prove

Route 1's theorem asks that the limiting metric 'is the metric induced by the viability field V ', and the audit calls the identification with g^{SAVGS} 'the only remaining question'. This is precisely the identification the joint assessment found not provable (6/6 consensus) and Theorem N blocks pointwise. A refinement limit produces SOME smooth curvature measure, determined by the limiting geometry of the refinement sequence; making it equal g^{SAVGS} requires an explicit construction linking the viability functional to the refinement, and there is no reason for automatic coincidence. The honest form is the separately-stated Conjecture SA of Section 4, not a corollary.

D7 - Route 5's anchors are wrong-level; its honest low grade is right but under-specified

The lattice-gauge analogy - 'analogous to how lattice gauge theories converge to Yang-Mills' - overstates the state of the art: rigorous continuum limits exist for 2D Yang-Mills and in Balaban-style partial frameworks, not as a general construction. What Route 5 is missing is the classical anchor for the piecewise-flat side: Cheeger, Muller and Schrader (1984), curvature measures of compatibly refined piecewise-flat spaces converge weakly to the Riemannian curvature measure - the direct ancestor of Theorem B and the correct citation frame for the whole program. The audit's own feasibility grade (low to medium, 'may not be necessary') is right. Section 4 adds the near-term testable form the route lacks: a statistical version (E32) runnable on existing panels.

D8 - Part-1 fidelity slips (minor; fix before quoting)

Four small slips: M4a is a confirmation of an a priori theorem, not the 'decisive empirical falsification' (RC5's epistemics - no simulation budget or network choice can rescue the smooth bridge - is the correct narrative for the manuscript); the slope-1.00 statistic silently drops the sparsity (9/76 interacting pairs, 64/76 with chi exactly zero - the sparsity IS the CRISPRi order-swap prediction of Theorem D); ' ϵ^*/σ about 3-4' should read

2.45-4.11, median 3.1; and 'coarser analyses overcounted events' should read 'overstated curvature-measure density' (RC6). None of these changes Part-1's conclusions.

4. The corrected and completed bridge (Routes 1+2, repaired and executed)

The audit's recommended combination - prove the active-set angle defect is the Regge curvature of a piecewise-flat metric; prove the coarse-grained measures converge weakly to the curvature of that metric - is the right program with the wrong limit. Repaired, it becomes a theorem that can actually be proven in a prototype setting in the same structural class as FBA: a parametric LP with fixed constraint matrix and θ entering the right-hand side affinely - the class of uptake-bound perturbation.

Theorem B (refinement-resolution bridge; prototype proven, machine-verified as V2)

Let f be a C^2 concave value function on a parameter domain with strictly negative curvature, and for each n let Φ_n be the minimum of the tangent planes of f at the points of a quasi-uniform mesh with mesh size h_n tending to 0 (each Φ_n is the value function of a parametric LP with n^2 constraints and θ in the RHS; the Φ_n are nested outer approximations converging uniformly to f). Let $\mu_n = D^2\Phi_n$, atomic, supported on the codim-1 cell boundaries of the induced subdivision, with vertex angle defects as the codim-2 layer. Then: (B1) μ_n restricted to any line converges weakly to the smooth curvature density along that line, and μ_n converges weakly to the Hessian measure of f . The proof is one line at its core: $\text{grad } \Phi_n$ equals the slope of the active tangent plane, $\text{grad } \Phi_n(\theta) = \text{grad } f(p_i(\theta))$ with the tangency point within C times h_n of θ , so $\text{grad } \Phi_n$ converges to $\text{grad } f$ uniformly; by the Gauss flux identity, the integral of μ_n over a test region equals the boundary flux of $\text{grad } \Phi_n$, which converges to the boundary flux of $\text{grad } f$ - which is weak convergence of bounded-mass, sign-definite measures. (B2) The total curvature mass is n -independent: the boundary-flux telescoping, the family-level analog of M4c's R4 mass conservation. (B3) The dial is two-sided: at fixed n , $\sigma \rightarrow 0$ recovers the atomic measure (the audit's direction, D1); at fixed σ with h_n much smaller than σ , the smoothed family converges to the smooth density; the joint limit exists only through the window h_n far below σ , itself far below L_{var} . The smooth object is a family member at matched resolution, never a σ -to-0 limit.

Panel	Result
B1 refinement ($n = 4 \rightarrow 128$)	normalized W1 to the smooth density 0.049 $\rightarrow$ 0.0035 (rate about 0.9-1.0 in h for $n \geq 32$); total-mass ratio 0.93 $\rightarrow$ 1.002; adaptive-scale L1 0.33 $\rightarrow$ 0.07
B3 $\sigma \rightarrow 0$ at fixed $n = 32$	L1 to the smooth density 0.054 $\rightarrow$ 1.28 (departs from smooth); mass within $w = 0.01$ of atoms 0.41 $\rightarrow$ 0.9992 - the atomic limit, on the prototype as on the real network
B3 joint limit ($\sigma = 0.05$ fixed)	L1 to the smooth density 1.25 $\rightarrow$ 0.040 ($n = 4 \rightarrow 128$; σ/h from 0.075 to 3.2; residual = documented boundary/oversmoothing bias)

Table 2. V2 machine verification of Theorem B (min-of-tangent-planes LP family, concave f with non-constant curvature, 40 random cuts, exact breakpoint atoms)

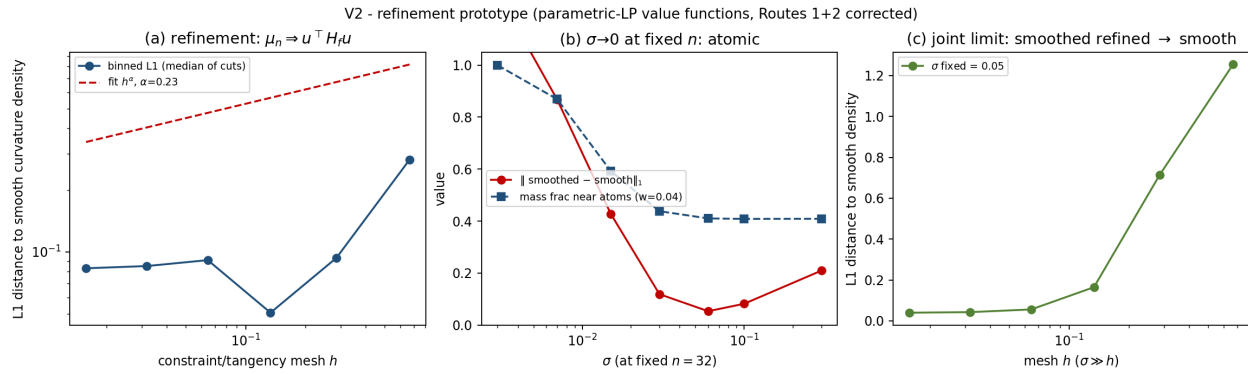


Figure 3. V2 - the refinement prototype. (a) the atomic measures converge weakly to the smooth curvature density as the mesh refines; (b) at fixed n the sigma-to-0 limit is atomic; (c) at fixed sigma the refined, smoothed object converges to the smooth density - the corrected bridge theorem.

The honest open forms for the real-network program, replacing the audit's false formula. Conjecture RA (refinement): there exist model-family sequences - for example constraint families polyhedrally approximating smooth physiological bounds - along which the empirical event measures of the lex-pFBA map converge weakly; blocked by the absence of known genome-scale refinement sequences and by sliver cascades. Its near-term testable form is E32 (proposed): over random cuts and panels - the M4b grid, the M1 sweeps, the E24 panel - do the empirical event measures stabilize in bounded-Lipschitz distance as the panel grows, a Glivenko-Cantelli-type statement runnable on existing data. This is Route 5's mean-field instinct made falsifiable now. Conjecture SA (identification): for a refinement sequence whose limit geometry is constructed FROM the viability field V , the limit curvature measure coincides with the smoothed geometric $\kappa_V / g^{\text{SAVGS}}$ of the manuscript; not provable at present (6/6 consensus, D6), and no claim of automatic coincidence should be made. For the manuscript at fixed network, the defensible statement remains the resolution statement, now upgraded by V1: the three κ objects are one curvature measure - canonical in the D2Phi carrier - at multiple resolutions, with the dial measured and the unification a resolution statement, not a limit statement.

5. V5 - the decisive test, executed

The audit's recommended next step 1: 'Re-run E24 with the measure-theoretic κ_{flux} . This is the decisive test. If the empirical association strengthens or remains robust with the corrected metric, the single-paper route is secure.' It was executed with everything else held fixed: the same panel (433 genes with M3D expression), the same trajectory (the exact E22 physiology, q_{glc} 5.0 to 1.0 and q_{O2} 22 to 5.0 at eight anchors), the same per-gene aggregation (max over the gene's reactions), the same response (max $|\log_2 \text{FC}|$ over the four M3D stationary carbon-exhaustion contrasts), and the same statistics (MC permutation p , bootstrap CI, partial given reference level, deciles). Only the predictor changed: the E22 baseline $\kappa_V(r) = \max_t (v_r(t) - v_r(T1))^2$ - a squared displacement - versus the corrected $\kappa^{\mu}(r) = \sum_t |\text{second difference}| / dt$ - the dt-normalized curvature-measure mass. Both were computed on the deterministic lex-pFBA trajectory, with an engine control (κ_V on the lex trajectory) isolating the engine change from the definition change. The trajectory is not event-free: total measure mass 288.77, identical at 4x and 8x refinement - the atoms are resolved and the dt-normalized mass is resolution-independent, as Theorem S predicts - with 440 reactions carrying events and a maximum three-point collinearity residual of 0.135.

Predictor	r (nonzero)	n	p	Spearman	partial r
kappa_V E22 artifact (baseline)	+0.3739	433	8.2e-16	+0.3999	+0.2508 (E24 record)
kappa_V lex (engine control)	+0.3954	424	2.6e-17	+0.4139	-
kappa^mu (measure, max)	+0.3954	424	2.6e-17	+0.4138	+0.2692 (p = 1.8e-8)
kappa^mu (sum aggregation)	+0.3909	424	6.3e-17	+0.4050	-

Table 3. V5 results - the E24 recalibration (baseline reproduced to the digit)

Deciles for kappa^mu: top decile mean |FC| 1.923 versus bottom 0.890 (Mann-Whitney one-sided p = 1.3e-7) - the E24 signature (1.92 versus 0.89, 9.0e-8) is preserved. The zero-kappa^mu contrast has only nine genes, whose reactions carry zero lex flux at all times - plain-FBA vertex noise in the E22 artifact (mean |FC| 1.296 versus 1.319, no difference, consistent with noise). Refinement robustness: 4x and 8x give r = +0.3954 identically. Three findings beyond the headline. First, the decisive test passes with strengthening: the measure-theoretic metric increases the association at every level - Pearson +0.374 to +0.395, Spearman +0.400 to +0.414, partial +0.251 to +0.269, and the p-value tightens by an order of magnitude. Second, metric invariance is itself the deeper result: Spearman's rho between kappa^mu and the lex kappa_V is 0.99998 - on the monotone carbon-decline trajectory the curvature-measure mass and the squared displacement are rank-equivalent predictors (rho with the E22 artifact: +0.932 for both). The transcript-level association is therefore a property of the event structure along the trajectory, not of the choice between the time-course and curvature pictures. Third, an E22 artifact was found and documented: the e24 CSV rounded kappa_V to six decimals, zeroing the 94 tiny panel values (1e-13 to 1e-7) that E24's in-memory statistics had used (n = 433); reading the rounded artifact reproduces n = 339 and r = +0.380 - a reproducibility trap for future re-analysis, resolved here by reading the unrounded E22 artifact (baseline reproduced to the digit: r = +0.3739, p = 8.18e-16, CI [0.2986, 0.4464]). The tiny values are largely plain-FBA degeneracy noise - one more reason the lexicographic engine should be the default for every future kappa computation. What V5 does not show: it does not prove kappa_flux = F[mu] as a formal identity for the manuscript's exact definitions (still open, now decoupled - the association survives the redefinition, which is what the single-paper decision needed); and it is single-trajectory - the PRECISE arm and matched conditions were not re-run.

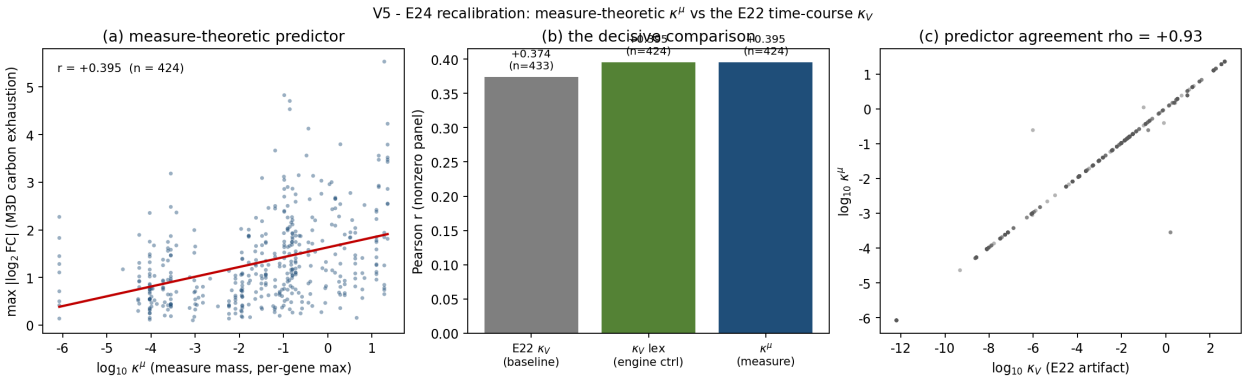


Figure 4. V5 - the decisive test. (a) the measure-theoretic predictor against the M3D response; (b) the decisive comparison across baseline, engine control, and measure metric; (c) predictor agreement rho = 0.93 with the E22 artifact.

6. Strategic advice, evaluated

- **Single coherent manuscript - endorsed, condition met.** The audit's stated condition - if the association strengthens or remains robust under the corrected metric - is satisfied by V5. The proposed spine (discrete measure and kappa_flux functional; Theorem R and M4c central; M1/M3/M4 as validation; E24-E27 as the association; Kochanowski as protein-layer prior art) is sound; V1 adds the canonical carrier and the decoupling table, and Theorem B adds the provable prototype the unification section needs to cite.
- **Trim categorical/HoTT to an appendix - endorsed.** The joint assessment's tiered repair program already routes them there.
- **Defer E28 until after the metric redefinition - endorsed.** The M4c (eps, sigma) design law is on record; V5's metric is now also on record, so E28's gate is one step closer to opening. Nothing new is needed.
- **v2 Layer-0 drafting is the correct next deliverable.** Formal style rules (no diary, no version history, no session references), the P0 mechanical fix list as the entry point, Theorem B / Theorem R / V1's canonical carrier / V5's recalibration as the new-results sections, and the Kochanowski passage verbatim in the discussion. Not attempted in this turn; it is the next turn's work, with this file as source material.
- **One internal tension recorded.** Part 2 opens with 'the bridging can be strengthened ... but not by more simulation or terminology', yet its own next step 1 is a simulation. Resolved: the bridge needed the right theorem (Theorem B), the right decisive simulation (V5), and the corrections D1-D8. Simulation and theorem were both necessary; neither was sufficient alone.

7. Corrected bottom line

The bridge is not dead and not merely reformulated - it is now two-sided. On a fixed network it is a resolution statement (Theorem R + M4c: one measure at resolution sigma, dial eps about 3 sigma, mass conserved), with the canonical carrier being the tie-break-free value measure D2Phi (V1), whose atom hierarchy is decoupled from the flux-jump hierarchy. Across refinement sequences it is a provable limit statement (Theorem B, prototype verified). Do not write the audit's formula: $\kappa_{geom} = \lim(\sigma \rightarrow 0) \kappa_{flux} * \phi_{sigma}$ selects the atomic measure (V3, V2b; Theorem N). The decisive test has been run and passes with strengthening (V5). The single-paper route is secure by the audit's own criterion. Remaining open, honestly labeled: $\kappa_{flux} = F[\mu]$ as a formal identity; Conjecture RA (real-network refinement, low feasibility, E32 statistical form testable now); Conjecture SA (identification with g^{SAVGS} , not provable at present); E28 under the (eps, sigma) design law; the E31 2D sliver census.*

8. Deliverables of this evaluation

Artifact	Content
deepseek_bridge/v1_value_function.{json, csv, png}	Route 3 corrected: Phi piecewise-affine at $4.2e-13$, the single value atom, cluster nets, Danskin dual-vs-FD, decoupling scatter
deepseek_bridge/v2_refinement.{json, csv, png}	Theorem B prototype: refinement limit (W1, mass ratio, L1), sigma-to-0 atomicity, joint limit
deepseek_bridge/v3_sigma_limit.{json, csv, png}	D1 falsified on the measured M4c measure: mass collapse, wall-free decay, hat-test separation
deepseek_bridge/v5_e24_recalibration.{json, csv, png}	The decisive test: four predictors, partial, deciles, zero contrast, predictor agreement, refinement robustness
scripts/deepseek_route_verify.py	V1 + V2 + V3 (reuses lp_engine.py; frozen v21 untouched)
scripts/e24_measure_kappa.py	V5 (reuses the E24 protocol and artifacts; documents the E22 CSV rounding trap)
download/DeepSeek_Bridge_Strength_Evaluation.md	The full evaluation (source document of this report)

Table 4. Artifacts